

RC circuit theory.

The nominal resistance is 10000 ohms.

The nominal capacitance is 470 microfarads.

The nominal time constant is 4.7 seconds.

The time constant equals resistance multiplied by capacitance.

Discharge model.

Voltage decreases exponentially during discharge.

At one time constant the voltage is approximately 36.8 percent of its initial value.

The initial voltage is 5.0 volts.

The logarithmic voltage slope is negative one divided by the time constant.

Uncertainty.

Resistance tolerance is 1 percent.

Capacitance tolerance is 5 percent.

The nominal tolerance interval is 4.42035 to 4.98435 seconds.

Component tolerances do not describe a confidence interval for fitted measurements.

Limits.

The ideal RC model assumes constant resistance and capacitance.

A log-linear fit can be biased by additive voltage noise.

Room humidity was not recorded.